Linux Commands Guide

How to read this guide?

1. Text written in red color are supposed to be user inputs. Replace it with your input.

2. Text writen in [square bracket] are the options available for the command. Different options work differently. Read manual to know meaning of options.

3. Text written in blue color shall not be used as it is. You have to replace it with appropriate value based on the description provided.

4. For privilege commands you need superuser permissions. Add sudo as prefix to your command to execute command as superuser.

Command	Purpose	Examples	Comments
ENVIRONMENT			
pwd	Shows current working directory	$ pwd	You can also get current directory with $PWD variable
man command	Shows manual/help page for the command	$ man ls	Shows ls command help
echo "Text"	Displays the text on console	$ echo "Hi There"	Prints Hi There on console
|	Pipe the output of commands	$ ls | grep txt	Filter the files having txt in their name in current directory
&&	Executes multiple commands only if previous command is successful	$ bash script.sh && echo "Successful"	Will print "Successful" only if script.sh executes successfully (with exit status 0)
sudo	Run commands with superuser	$sudo mkdir /home/newuser	Gets the superuser permissions for given command
sudo su	Change to superuser	$ sudo su	Gets the root user access. exit if you want to go back to normal user permissions.
which command	Get location of command	$ which ls	Shows location where it found the command. Traverse through all locations from $PATH variable value

export NAME=VALUE	Export the environment variable	$ exort JAVA_HOME=/usr/lib/jdk-1.8/	If you exit the shell environment variables goes away. If you want environment variables to be always set, add this command to .bash_profile or .bashrc or /etc/profile file
env	List all environment variables	$ env	
$PATH	Path variable for searching commands location	$ export PATH=$PATH:/new/location	This will add /new/location to PATH variable so that commands from /new/location can be executed without providing absolute path
$?	Holds exit status of last command executed	$ echo $?	Displays exit status (numeric) of whichever command you had executed before executing this command. Exit status 0 mean successful. Non 0 exit status generally means failure.

~	Home directory (similar to $HOME)	$ cd ~	Takes to home directory of user executing the command
$USER	Current user name	echo $USER	Prints current username
.	Current Directory	$ cp /tmp/file .	Copies /tmp/file into current directory
..	Parent directory	$ cd ..	Moves directory to parent directory
<TAB>	Command auto-completion		You don't need to type complete command. Just type initial characters and press tab.
FILESYSTEM			
ls [-l, -a, -h]	List the current directory contents	$ ls -l	List all files with detailed columns permissions, size etc
cd dir	Change directory	$ cd /tmp	Changes directory to /tmp
mkdir dir	Create new directory	$ mkdir mydir	To create directory at different location use absolute path e.g /home/chetan/mydir
mkdir -p /dir1/dir2/dir3	Create directories recursively	$ mkdir /tmp/dir1/dir2	No need to create child directories separately
rm file	Removes file	$ rm myfile.txt	
rm -f file	Removes file forcefully.	$ rm -f myfile.txt	Be careful with this command
rm -rf dir	Removes directory recursively	$ rm -rf mydir	Be extremely careful with this command

cp file1 file2	Copy file1 as file2	$ cp test1 test2	Makes exact copy of file
cp -r dir1 dir2	Copy dir1 as dir2 recursively	$ cp -r dir1 dir2	
mv file1 file2	Rename file1 as file2	$ mv test1 test2	
mv file1 file2 file3 dir1	Moves file1, file2, file3 inside dir1	$ mv file1 file2 /tmp/	Move both files to /tmp directory
cat file	Display complete contents of file on console	$ cat myfile.txt	Displays file content on the console
less file	Show part of file in page mode		
tail file	Show last 10 lines of the file		
tail -f file	Continuously show output of file		Useful while looking at log files while being written by a process in real time

chmod ugo file	Change file permissions. u is the user's permissions g is the group's permissions o is everyone else's permissions.
	Value of u,g,o is between 0 to 7
	7 — full permissions
	6 — read and write only
	5 — read and execute only
	4 — read only
	3 — write and execute only
	2 — write only
	1 — execute only
	0 — no permissions
		$ chmod 600 myfile
		$ chmod 400 myfile
		$ chmod 755 directory
			Grant only user read/write permissions on file
			Grant only user read permissions on file
			Grant User full permissions. Group/Others Read/Execute permissions on the directory.

vi file	Open file for editing using vi edior	$ vi myfile.txt	Look at section below for useful vi commands
> file	Redirect standard output to new file	$ date > datefile.txt	Writes the date command output to datefile.txt.
>> file	Append the output to existing file (or creates new file)	$ date >> datefile.txt	Does not overwrite file contents. Appends the output to existing file. If file does not exist, it will create the file.
history	Shows the command history		History is typically stored in ~/.bash_history file

CTRL+R and type command initials	Searches command you recently executed		Useful if you want to re-execute same command
USEFUL COMMANDS PIPED TOGETHER			
ps -ef | grep SSH	Only list SSH processes		
netstat -an |grep :80	Filter port 80 port related output		
command | cut -d : -f2	Cut the output of the command by spaces and print 2nd word		
SYSTEM			
uname -a	Display Linux system information	$ uname -a	
cat /etc/redhat-release	Show which version of redhat installed		
hostname	Displays the system host name	$ hostname	
hostname -i	Displays host IP address	$ hostname -i	
shutdown	Shuts down the system	$shutdown 0	Shuts down the system immediately
reboot	Reboots the system	$ reboot 0	
date	Shows current date time	$ date	
whoami	Shows who you are logged in as	$ whoami	
df [-h]	Shows filesystem space usage	$ df -h	Displays the disk space in MB, GB format
du [-h] dir	Shows directory spaec usage	$ du -h /home	Displays disk usage of /home directory
free	Shows free memory and swap space	$ free	Displays free memory (RAM) and SWAP
top	Shows CPU, Memory usage	$ top	Press q to exit the display

ps	List running processes	$ ps -ef	List all processes
kill -9 PID	Send SIGKILL signal to process	$ kill -9 189	Ends the process identified by process id 189
service servicename start	Starts the service	$ service httpd start	Starts httpd service which listens on port 80 by default
service servicename stop	Stops the service	$ service httpd stop	
service servicename status	Status of the service	$ service httpd status	Shows if service is running or stopped
systemctl start servicename.servi	Start the service	$ systemctl start httpd.service	Newer version of service
systemctl stopc servicename.servi	Stops the service	$ systemctl stop httpd.service	
systemctl list-unit-files --state=ena	Shows status of all services	$ systemctl list-unit-files -state=enabled	List all active processes
NETWORKING			
wget URL	Download file from the internet	$ wget http://example.com/file_url	Used to download packages/images etc
curl URL	Hit the HTTP/HTTPS	$ curl http://example.com/endpoint	May be used to hit the REST API
ssh user@host [-i key]	SSH to remote host	$ ssh ec2-user@10.100.0.24	
scp file user@host:/dir	Copy file to remote host in remote dir	$ scp myfile.txt ec2-user@10.100.0.24:/home/chetan/	Copied local file to remote host in /dir on remote host
scp user@host:/dir1/file dir2	Copy file from remote host to current host	$ scp ec2-user@10.100.0.24:/home/chetan/myfile.txt myfile.txt	Copies myfile.txt from remote host to local directory
ping host	Ping host and check connectivity		Must have ICMP enabled on host
netstat -an	List all listening ports on the host		Useful white troubleshooting network issues. Verify that port is actually listening. Should be run on server.
lsof-i:port	List process listening on port	$ lsof -i:33034	netstat does not show which process is listening on given port. Should be run on server.
nc -z host port -v -w5	Check the network connectivity to given host and port	$nc -z 10.44.22.11 22 -v -w5	Check whether you can SSH on port 22 on target host. Very useful to troubleshoot network issues. Should be run from client machine.

FILE AND TEXT SEARCHING			
grep somestring file	Display all the matching pattern lines in the file	$grep ERROR file	somestring can also be regular expression. Based on grep version you might have to use -E option to enable regex search. Newer systems may have egrep command instead of grep.
grep -r somestring directory	Recursively search all files in the directory for matching pattern	$grep -r ERROR dir	Output contains filenames:matching line

grep -i somestring file	Case insensitive search	$grep -i ERROR file	Searches for both error and ERROR string pattern
find filename	Finds all files with filename from the current directory	$find myfile.txt	Lists all files with name filename recursively from current directory
find /dir -name filename -type f	Finds all regular files with filename in /dir path	$ find / -name myfile.txt -type f	Finds myfile.txt in complete system
sed -e 's/search/replace/g' file	Search and replace pattern in the file. Does not modify file contents.	$ sed -e 's/chetan/CHETAN/gc' employees.csv	This does not modify file contents. Just displays output on the console
sed -i -e 's/search/replace/g' file	In place replace file contents	$ sed -i.bak -e 's/chetan/CHETAN/gc' employees.csv	Modifies existing file. If you want to backup original file, then include -i.bak option which creates backup file like employees.csv.bak
cut -d " " -f1	Cuts the output by space and prints first word	$ echo "Hi There" | cut -d " " -f1	Print only "Hi". Cut command will always be after | (pipe) so that it can work on command output
PACKAGE INSTALLATIONS			
Linux Variants (RedHat, CentOS, Fedora)			
yum update	Update your yum repository	$yum update	You may need to use **sudo** if you are running command as non-priviledged user
yum search package	Search for package in configured yum repositories	$yum search mysql	
yum install package	Installs the package on the system	$yum install httpd	
yum info package	Display package details (Version, Repo etc)	$ yum info mysql	
yum remove package	Uninstall package from the system	$ yum remove httpd	Removes the package from the system

rpm -ivh packagefile	Install the package	$ rpm -ivh mozilla-mail-1.7.5-17.i586.rpm	You need to first download the package file using say wget or curl command.
rpm -ev package	Erase/remove/ an installed package	$ rpm -ev mozilla-mail	Removes the package from the system
Debian Variants (Ubuntu, Kali Linux)			
apt-get install package	Installs the package on the system	$ apt-get install apache2	
apt install package	Installs the package on the system	$ apt install apache2	Latest version of apt-get
apt remove package	Removes package on the system	$ apt remove apache2	Removes package
apt search package	Searches package in apt repository	$ apt search mysql	
dpkg -i packagefile	Install the package	$ dpkg -i zip_2.31-3_i386.deb	
dpkg -r package	Remove/Delete an installed package except configuration files	$ dpkg -r zip	
COMPRESSION			
tar cf file.tar file1 file2 file3	Creates a tar named file.tar containing files	$ tar cf file.tar *	Adds all files in current directory to file.tar
tar xf file.tar	Extracts tar file into current directory	$ tar xv file.tar	

tar cfz file.tar.gz file1 file 2 file3	Creates tar file and compresses it with GZIP	$ tar cfz file.tar.gz *	
tar xzf file.tar.gz	Unzips and extracts all files from file.tar.gz	$ tar xzf file.tar.gz	
tar xf file.tar -C dir	Extract tar file into dir instead of current directory	$ tar xf file.tar -C /tmp	Extracts all files under /tmp directory
vi or vim			
vi file	Open file for editing	$ vi myfile.txt	You must have permissions to create file

i	Get into insert mode		So that you can write the contents. Note that only while typing contents or for pasting (right click) contents you will be in insert mode. Otherwise you will always be in ESC mode.
<right click>	Paste the contents copied into the file		You must be in Insert mode
esc	Exit the insert mode		So that we can execute below all vi commands
:w	Save file		
:wq	Save file and exit vi		
:x	Save file and exit vi		
:q	Exit without making any changes to file		
:q!	Forcefully exit without making any changes to file		
:set nu	Set line numbers in the file		Just for Display. Lines numbers are not added to file contents.
:set ic	Set ignore case feature		Used while searching through the file in case-insesitive way
yy	Copy or yank the a line		
p	Paste the copied line		
u	Undo the changes		Undo the changes made. Press u multiple times if you want to undo more changes.
^	Move cursor to start of the line		
$	Move cursor to end of the line		
%	Go to matching bracket		If your cursor is at { then typin % will take you to matching }

x	Delete single character at the cursor		
dd	Delete complete line		
dw	Delete complete word		
5dd	Delete 5 lines		
/string	Searches string in the file	/name	Press "n" to go on next occurance of the string

:%s/pattern/replace/gc	Search and replace string in file	:%s/name/Name/gc	"c" asks for confirmation. Without "c" it will replace globally.
VI EDITING FLOW			
1. Open file using command: vi *filename*			
2. Press i so that you get into Insert mode			
3. Type or paste whatever you want to write			
4. Press ESC to get out of insert mode			
5. Run vi commands like x for deleting some character, dd to delete line or more commands			
6. Go back and forth into insert mode (pressing i) or esc mode (pressing esc) as per your need			
7. While in Esc mode press :wq to save the file and quit vi editor			
VI TROUBLESHOOTING TIPS			
1. The CAPSLOCK enables different commands. Make sure its not ON accidently.			
2. Always have practice to press ESC key so that you can run the vi commands			

3. If things are not working, press ESC and quit by pressing :q!